Contents

Abstract

This project introduces an energy-efficient single-ended 10T SRAM design using adiabatic positive feedback. The design integrates a built-in read-assist and power-gating techniques to enhance read stability and writability, respectively. By utilizing single-ended read/write operations and transistor stacking, the design achieves reduced power consumption. Simulation in 18-nm CMOS technology at 3V demonstrates proposed design consumes significantly less power compared to Existing designs. Overall, this SE10T SRAM design presents a promising solution for low-power system-on-chip applications, maintaining read stability and writability while minimizing energy consumption.

Introduction

- Modern System on chip applications such as IOT and portable devices require low-energy systems for extended operation.
- Near-threshold voltage (Vth) designs offer reduced power consumption while maintaining operational efficiency.
- Designing stable circuits resilient to process, voltage, and temperature variations presents significant challenges.
- Static random access memories (SRAMs) dominate system on chip energy consumption and require optimization.

- Conventional 6T SRAM cells suffer from issues like degraded SNMs, read/write conflicts, high power consumption, and half-select problems.
- Proposed solutions involve modifying 6T SRAM cell structures and implementing new techniques to improve stability, writability, and reduce power consumption and half-select issues.

Motivation

The main motivation behind developing an energy-efficient single-ended read/write 10T near-threshold SRAM is to reduce power consumption.

Literature Review

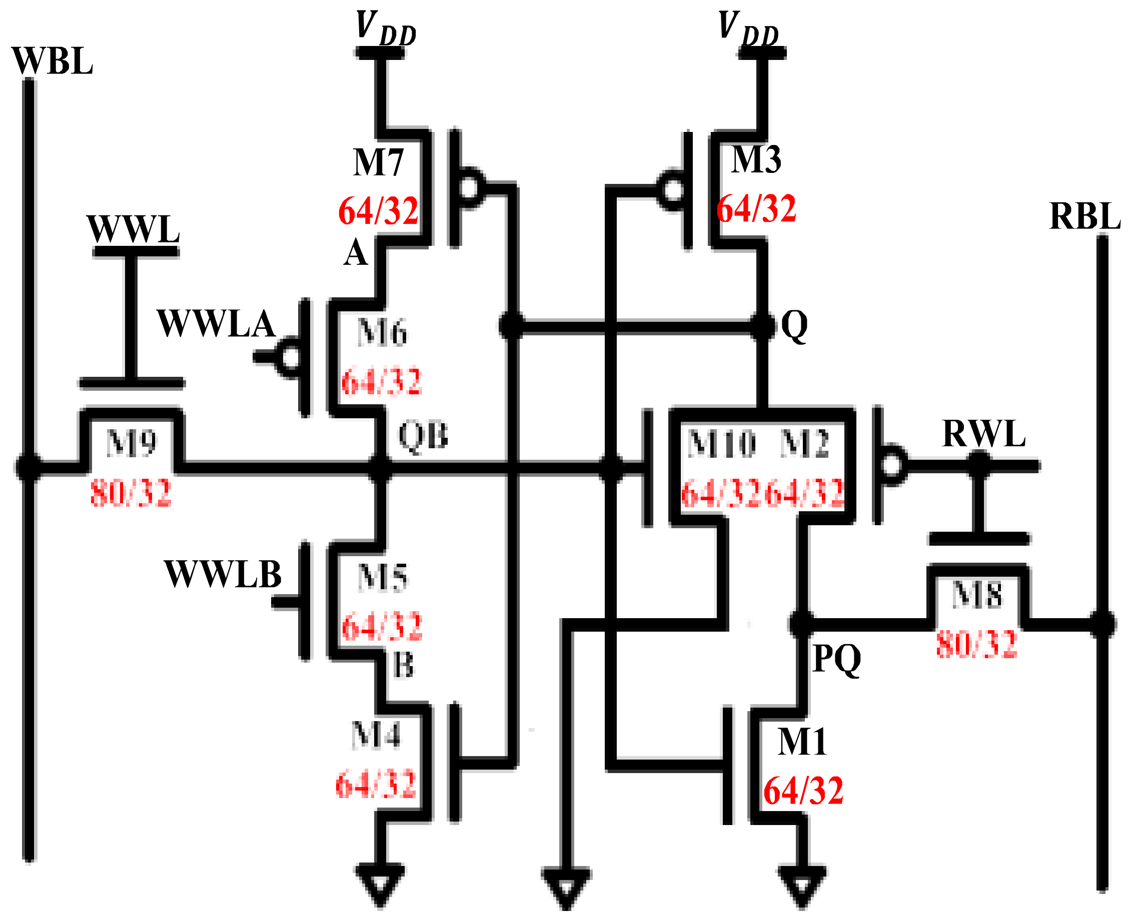
S.No	Reference No	Title	Year of published	Merits
1	[1]	A 40 nm 512 kb cross-point 8 T pipeline SRAM with binary word-line boosting control, ripple bit-line and adaptive data-aware write-assist.	Dec. 2014	Adaptive Data-Aware Write-Assist
2	[2]	Half-select free and bit-line sharing 9T SRAM for reliable supply voltage scaling.	Aug. 2017	Innovative SRAM Design
3	[3]	Pentavariate *V*min analysis of a sub threshold 10T SRAM bit cell with variation tolerant write and divided bit-line read.	Oct. 2018.	Improved Variation Tolerance
4	[4]	A write bit-line free sub-threshold SRAM cell with fully half-select free feature and high reliability for ultra-low power applications.	Feb. 2022	Sub-threshold Operation

Problem Statement

The problem this project addresses developing an energy-efficient single-ended read/write 10T near threshold SRAM necessitates optimizing circuitry to balance power consumption with reliability, particularly when operating transistors near their threshold voltage.

Existing methodology

- **Near-Threshold Operation**: Operating SRAM near its threshold voltage reduces energy consumption while maintaining functionality, requiring careful design to ensure stability.
- **10T SRAM Cell Design**: The SRAM cell design with 10 transistors per cell balances factors like speed, area efficiency, and energy consumption.
- **Read/Write Circuitry Design**: Efficient design of read/write circuits minimizes energy consumption during data access and storage.
- **Sense Amplifier Design**: Sense amplifiers detect and amplify small voltage differences during read operations efficiently.



Schematic of the existing SE10T SRAM cell along with its control signals status.

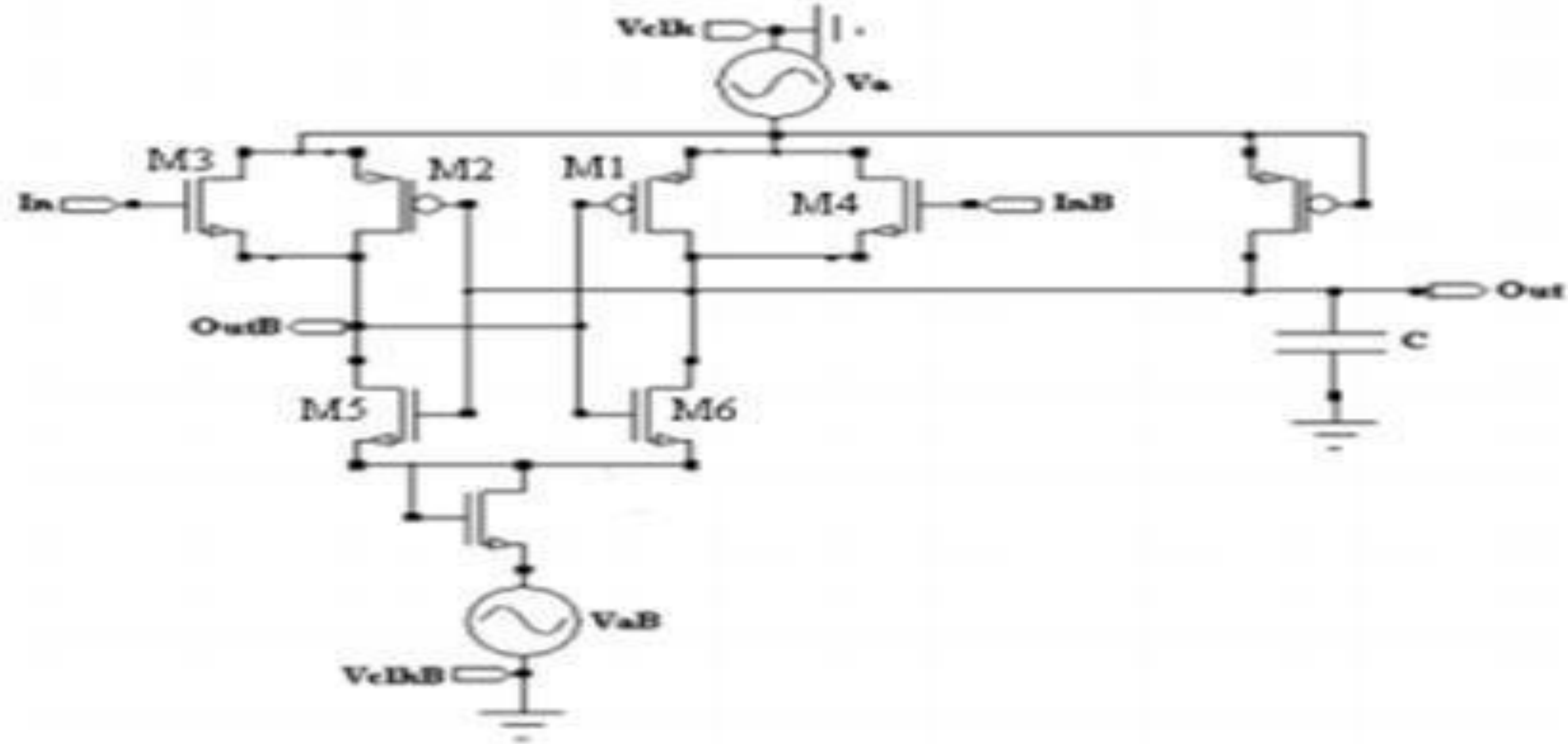
Proposed methodology

Adiabatic logic:

Adiabatic Logic is a way to save energy in electronic circuits, especially in smaller, faster ones like CMOS technology. It focuses on keeping energy levels stable by making sure voltages stay similar and current flows smoothly. Special designs and materials, like SCRL(Stage Charge Recovery Logic) and silicon nanowires, are used to make circuits more efficient and save power.

Positive Feedback Adiabatic Logic

In adiabatic logic circuits, positive feedback via cross-coupled inverters and NMOS transistors on a four-phase voltage clock improves energy efficiency and signal amplification. PFAL uses steady feedback signals for performance regulation, reducing power consumption and enhancing noise margin. This approach also enables energy recovery by recycling energy during logic transitions, making adiabatic logic more efficient than traditional CMOS logic.

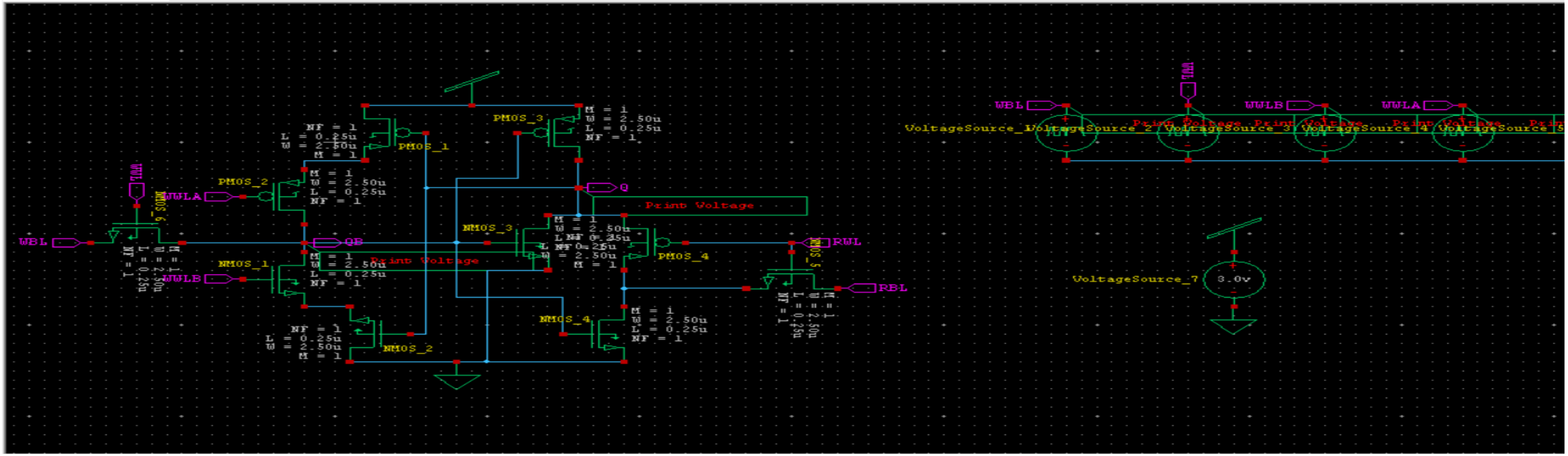


Schematic of the Proposed10T SRAM cell

Simulation Results

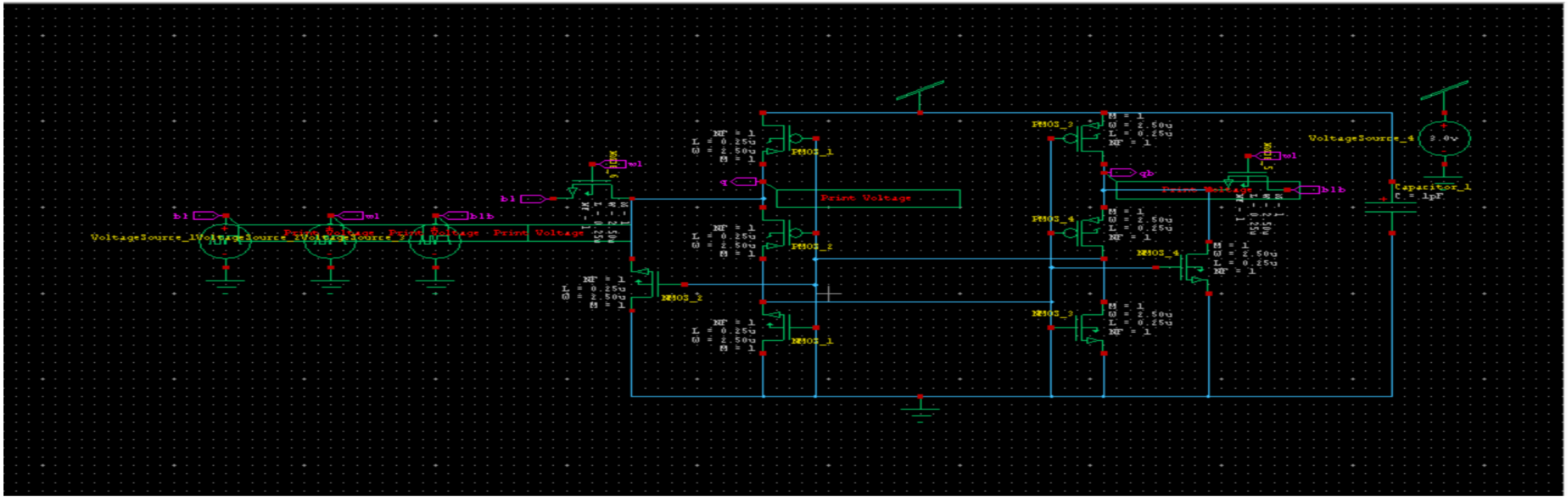
Schematic Circuit for Existing 10T SRAM

The Existing 10T SRAM cell is designed using 32nm CMOS technology and developed using Tanner EDA. The operating voltage for this cell is 5 volts.



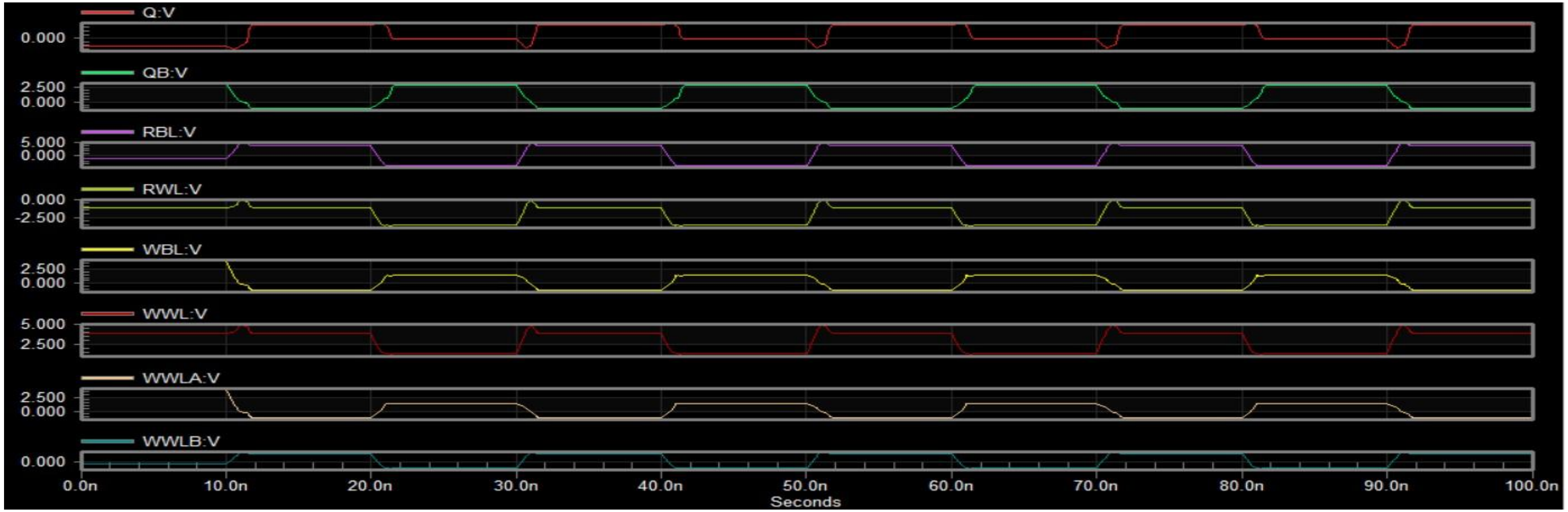
Schematic Circuit for Proposed 10T SRAM

The Proposed 10T SRAM cell is designed using 18nm CMOS technology and developed using Tanner EDA. The operating voltage for this cell is 3 volts.



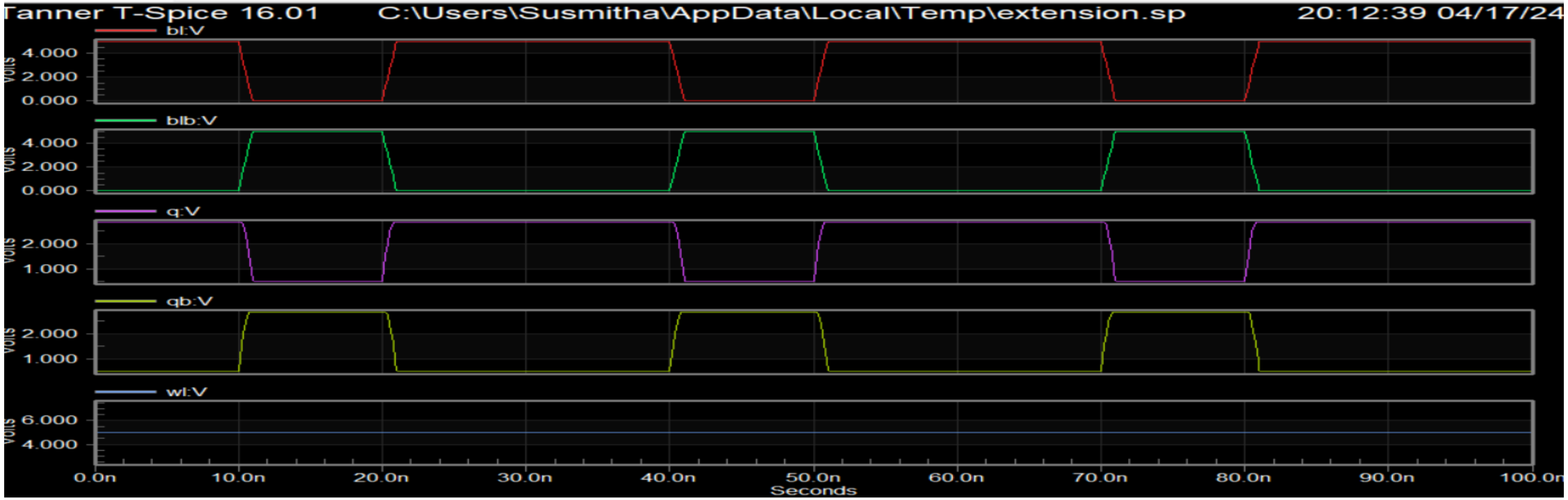
Waveforms for Existing 10T SRAM

For a read operation in the memory cell, the Read Bit Line (RBL) is set to 1, while the Write Bit Line (WBL) is set to 0. As a result, the output Q represents the read data from the memory cell, which is set to 1, while the complementary output QB represents the write data to the memory cell and is set to 0.



Waveforms for Proposed 10T SRAM

For a Write operation in the memory cell, BL is set to 1, while the BLB is set to 0. As a result, the output Q represents the write data to the memory cell, which is set to 1, while the complementary output QB represents the read data from the memory cell and is set to 0.



Power Results for Existing 10T SRAM

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Power Results

VVoltageSource_7 from time 0 to 100
Average power consumed -> 1.626605e-013 watts
Max power 2.905879e-003 at time 1.15105e-008
Min power 2.873311e-011 at time 0


Parsing                              0.00 seconds
Setup                                0.01 seconds
DC operating point                   0.06 seconds
Transient Analysis                   0.04 seconds
Overhead                             1.07 seconds
----------------------------------------------
Total                                1.19 seconds
```

Power Results for Proposed 10T SRAM

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Power Results

VVoltageSource_2 from time 0 to 100
Average power consumed -> 8.952503e-016 watts
Max power 1.737586e-004 at time 4.09191e-008
Min power 0.000000e+000 at time 0


Parsing                              0.01 seconds
Setup                                0.00 seconds
DC operating point                   0.08 seconds
Transient Analysis                   0.00 seconds
Overhead                             0.80 seconds
-----------------------------------------
Total                                0.90 seconds
```

Performance Metrics

Parameter	Existing 10T SRAM	Proposed 10T SRAM
Number of Nodes	12	11
Power (mWatts)	0.29	0.17
Delay (seconds)	1.19	0.9
Power Delay Product (Watts sec)	$3.4x10^{-1}$	$1.5x10^{-1}$

Conclusion

The Adiabatic Positive Feedback Method effectively enhances the energy efficiency of Single-Ended 10T Near-Threshold SRAM designs by leveraging adiabatic switching and positive feedback principles. This method significantly reduces energy consumption during read and write operations, achieving notable improvements over conventional SRAM designs

References

- A 40 nm 512 kb cross-point 8 T pipeline SRAM with binary word-line boosting control, ripple bit-line and adaptive data-aware write-assist.
- Half-select free and bit-line sharing 9T SRAM for reliable supply voltage scaling.
- Pentavariate *V*min analysis of a subthreshold 10T SRAM bit cell with variation tolerant write and divided bit-line read.
- A write bit-line free sub-threshold SRAM cell with fully half-select free feature and high reliability for ultra-low power applications.
- A highly stable low-energy 10T SRAM for near-threshold operation

THANK YOU